

Title of the Paper Goes Here: Capitalize the First Letter of Major Words, Centered, 14-Point Times New Roman, on the Second Line from the Top Margin, Preferably not More Than 3 Lines Long

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ABSTRACT

Use 8.5 x 11 paper size, with 1" margins on all sides. A required 200-250 word abstract starts on this line. Leave two blank lines before "ABSTRACT" and one after. Use 11 point Times New Roman here and single (11 point) spacing. The abstract is a very brief summary highlighting main accomplishments, what is new, and how it relates to the state-of-the-art.

KEYWORDS

List of at most five key words

1. INTRODUCTION

Paper starts here with two blank lines before first section title. Use 8.5 x 11 paper size, with 1" margins on all sides. All text should be "justified" – evenly distributed between the margins. Use one blank line before and after each subsequent section's title. Section titles have style "Heading 1", are **11 point font**, must be all uppercase and text aligned left, and must be numbered in Arabic numerals as shown above. Introduce the topic of your work in this section.

Do not indent the first line of a paragraph; rather add one blank line between paragraphs. Use Times New Roman font (and Symbol if needed) and **font size 11** for the main body of the paper. There are four types of reference styles: journal paper [1], proceeding paper [2], book [3], and website [4]. References to

* Footnote, if necessary, in Times New Roman font and font size 10

websites are discouraged, but acceptable if absolutely necessary. It is the author's responsibility to check links in the PDF file of your paper. All references should be cited in the text in numerical order, in order of appearance as [5-7]. Do not include bookmarks or hyperlinks to references, figures, and tables in the text of your paper in your final PDF document. Do not include highlighting, page numbers, headers, or footers.

Because the paper will be included as a .pdf file for the proceedings, the author will get the best results from Word or WordPerfect by using the Acrobat Distiller or Acrobat PDFWriter as the default printer. When creating the PDF version, check the "Embed All Fonts" option. Note that it is the author's responsibility to review the final PDF version of the paper to ensure proper translation into PDF. Final PDF file size should be no more than 4 MB. The recommended paper length is 8-10 pages. Papers will incur a publication fee of \$100 per page starting with p. 11; there is no charge for pp. 1-10. **The maximum allowable length is 14 pages. Papers with more than 14 pages will be rejected.**

2. SECOND OR SUBSEQUENT MAJOR HEADING (FONT SIZE 11 POINT)

A logical division of your paper into sections, etc., makes it so much easier to understand. The style for subsection titles and all text in this template is "Heading 2," "Heading 3," etc. All text in this template is "Body Text 3." Make sure to **avoid widow/orphan** lines.

2.1. Subsection Title: First Character of Each Non-Trivial Word is Uppercase (11 Point)

Add a blank line before and after secondary titles. Secondary titles should start flush left, and are numbered as illustrated above.

Equations should be centered and sequentially numbered to the flush right of the formula. Use Times New Roman or Euclid font with a font size 11 for the text variables and corresponding sizes proportionally for the subscripts and superscripts.

$$\partial_x U_k + \partial_x F_k(U) = \partial_x K U_k^{3/2} \partial_x U_k \quad (1)$$

The continuation of a paragraph after an equation is not indented. All paragraphs, section or subsection headings, as well as equations are separated by just one single empty line.

2.1.1. Sub-subsection level and lower: only first character uppercase (11 point)

Figures and tables should appear as closely as possible to where they are first cited, e.g., Fig. 1, in the text. Figures are numbered in Arabic numerals, with the caption centered below the figure, in **boldface**. Use two blank lines before the figure and after the figure caption.

When importing figures or any graphical image please verify two things:

- Any number, text or symbol is in Times New Roman font and is not smaller than 10 point after reduction to the actual window in your paper;
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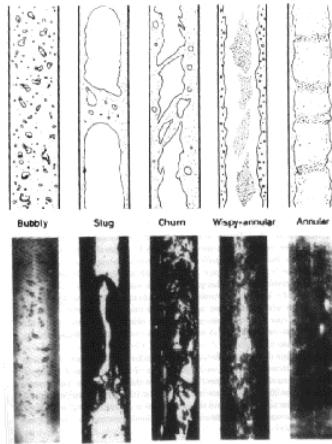


Figure 1. Sample Figure.

Tables, such as Table I, are numbered in Roman numerals, with the table title in **boldface** centered above the table with a blank line between the title and actual table. Use two blank lines before and after the table.

Table I. Sample table: accuracy of nodal and characteristic methods				
Mesh	8 x 8	16 x 16	32 x 32	64 x 64
Nodal	1.000×10^{-1}	2.500×10^{-2}	6.250×10^{-3}	1.563×10^{-3}
Characteristic	1.000×10^{-1}	2.500×10^{-2}	6.250×10^{-3}	1.563×10^{-3}

Use SI Units. Conventional (non-SI) quantities may follow parenthetically if the author desires.

3. CONCLUSIONS

Present your summary and conclusions here.

NOMENCLATURE (IF NEEDED)

If variables are extensively used in the text, a Nomenclature section would be helpful to the readers.

ACKNOWLEDGMENTS

Acknowledge the help of colleagues, and sources of funding, if you wish.

REFERENCES

List references here as a numbered list in left-aligned text, in order of their first use in the paper. Include the following information (as applicable). Please include DOIs wherever possible. Do not include these sentences in the full paper.

1. B.C. Author(s), "Title of the Journal Paper," *Journal Name in Italic*, **Volume in Bold** (followed by the issue number in parentheses if known, see Ref. 5 below), pp. 34-89 (2007).
<https://doi.org/10.987.6543.21>

2. D.E. Author(s), "Article Title," *Proceedings of Meeting in Italic*, Location, Dates of Meeting, Vol. 31, pp. 134-156 (2012).
3. F.G. Author, *Book Title in Italic*, Page or Chapter Numbers, Publisher, City and Country (1995).
4. "NEAMS Update," <http://www.ne.anl.gov/pdfs/NEAMS/NEAMSQuarterlyReport10-2014.pdf> (2014).
5. M. Ishii and S. Kim, "Development of One-group and Two-group Interfacial Area Transport Equation," *Nucl. Sci. Eng.* **146**(3), pp. 257-273 (2004). <https://doi.org/10.13182/NSE01-69>
6. R.P. Martin and C. Frepoli, "Physical Systems for Regulatory Investigation: Part II Evaluation Methodology," *Proceedings of 2014 International Topical Meeting on Advances in Thermal Hydraulics (ATH'14)*, Reno, Nevada, June 15–19, 2014 pp. 15-24 (2014).
7. S. Glasstone and W.H. Jordan, *Nuclear Power and Its Environmental Effects*, pp. 50, American Nuclear Society, La Grange Park, Illinois, USA (1980).

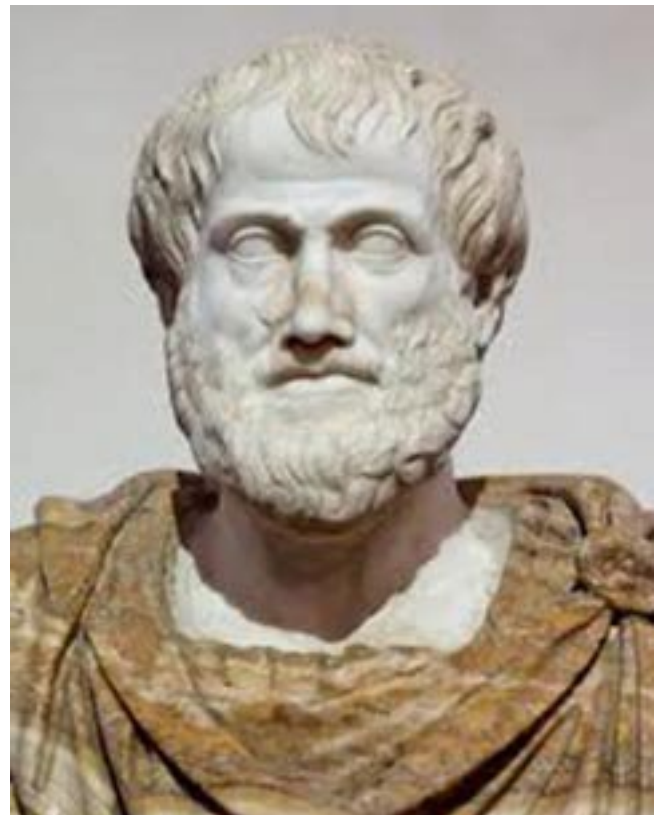
APPENDIX A

If necessary, include Appendices numbered in upper case alphabetical order.

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Aristotle

Biography Summary: Aristotle was a famous Greek philosopher who is said to have written about his theories and opinions in 150 philosophical treatises. Although only fragments of his work and lecture notes survived, we can establish that Aristotle had been able to make significant contributions towards many different subjects including physics and zoology, anatomy, embryology, geology, geography, astronomy, meteorology, literature and poetry.



Aristotle Fact File Biography: Lifespan: 384 BC – 322 BC.

~~Full Name: Aristotle was also known as 'the master of them that know' and 'the Philosopher'~~

~~Date of Birth: He was born in 384 BC~~

~~Place of Birth: Aristotle was born in Stagira, Chalcidice, Greece~~

~~Family background: His father was a doctor called Nicomachus~~

~~Education: Aristotle was educated at Plato's Academy Aristotle died in 322 BC~~

Aristotle Fact 1: ~~Aristotle was born in 384 BC in Stagira, Chalcidice, Greece. His name means 'the best purpose'.~~

Aristotle Fact 2: ~~He was a Greek philosopher and scientist who offered many profound views and theories. Aristotle is said to have written about many of his theories in 150 philosophical treatises, 30 of which survived.—~~

Aristotle Fact 3: ~~His father was a doctor called Nicomachus, he was King Amyntas of Macedon's personal physician and friend. We know little about Aristotle's childhood.—~~

Aristotle Fact 4: ~~Aristotle moved to Athens when he was approximately 18 years old to study at Plato's Academy. He was educated at the academy for twenty years. The institution was founded by Plato, a mathematician and philosopher in Classical Greece, it is also known as the Platonic Academy.—~~

Aristotle Fact 5: ~~Aristotle left Athens around 348 BC and traveled with another philosopher called Xenocrates to Asia Minor (modern-day Turkey). He was accompanied by Theophrastus to the Greek island of Lesbos where they studied zoology and botany.—~~

Aristotle Fact 6: ~~Aristotle married a Greek biologist known as Pythias the Elder, she was the adoptive daughter of Hermias of Atarneus. They had a daughter together whom they named Pythias.—~~

Aristotle Fact 7: ~~From 343 BC, he taught Alexander III of Macedon, known famously as Alexander the Great. Alexander's father, Philip II of Macedon, requested Aristotle to become his son's tutor. Aristotle~~

became the head of the royal academy of Macedon, an ancient kingdom of Classical Greece. He also taught two other future kings called Ptolemy and Cassander.—

Aristotle Fact 8: In 335 BC, he returned to Athens to establish his own school known as the Lyceum. He taught at the school for twelve years.

Aristotle Fact 9: Whilst Aristotle was teaching at the Lyceum School, his wife Pythias died. After her death, he began a relationship with Herpyllis of Stagira. They had a son together whom they named after Aristotle's father, Nicomachus.

Aristotle Fact 10: During his time at the Lyceum School in Athens, Aristotle studied many different subjects including physics and zoology, anatomy, embryology, geology, geography, astronomy, meteorology, literature, poetry. He contributed greatly towards many of these subjects, offering his theories and opinions on aesthetics, metaphysics, politics, ethics, economics, government, psychology, rhetoric and theology. He wrote numerous philosophical treatises, some of which survived. Some of his works are believed to be lecture notes for his students.

Aristotle Fact 11: The most significant and important philosophical treatises written by Aristotle include Physics, Metaphysics, Politics, De Anima: On the Soul, Nicomachean Ethics and Poetics

Aristotle Fact 12: Following the death of Alexander the Great, Aristotle was denounced by Eurymedon the Hierophant, the representative of Demetra of Eleusis, for not honoring the gods. As a result, Aristotle fled to Chalcis on the island of Euboea, Greece in 322 BC to avoid being sentenced to death.

Aristotle Fact 13: It wasn't until the Middle Ages that scholars rediscovered and became interested in his work and findings. He became known as 'The Master of Them that Know' and 'The Philosopher'.

Aristotle Fact 14: He believed that the universe was eternal, that there was neither a beginning nor an end. He accepted that things can change; however, he fundamentally believed that the overall conditions would always remain the same.

Aristotle Fact 15: He studied zoology very closely and through careful observation, he was able to produce his findings. His work in zoology contained some errors, however, much of what he reported was incredibly accurate.

Aristotle Fact 16: His careful study of animals enabled him to classify them into different groups and species. Although, much more developed today, he was able to split the animals into two major groups: animals with red blood and animals without red blood. His findings correspond closely with the distinctions later recognized between vertebrates and invertebrate animals.

Aristotle Fact 17: The following quote by Aristotle offers his opinion and theory on the Earth and the changes which he observed and predicted:

'The same parts of the earth are not always moist or dry, but they change according as rivers come into existence and dry up. And so the relation of land to sea changes too and a place does not always remain land or sea throughout all time, but where there was dry land there comes to be sea, and where there is now sea, there one day comes to be dry land. But we must suppose these changes to follow some order and cycle. The principle and cause of these changes is that the interior of the earth grows and decays, like the bodies of plants and animals...

But the whole vital process of the earth takes place so gradually and in periods of time which are so immense compared with the length of our life, that these changes are not observed, and before their course can be recorded from beginning to end whole nations perish and are destroyed.'

Aristotle Fact 18: He studied the science of the Earth and the weather, and wrote about his findings in a Meteorology treatise. He managed to work out the hydrologic cycle. The following quote by Aristotle describes his findings on the hydrologic cycle: 'Now the sun, moving as it does, sets up processes of change and becoming and decay, and by its agency the finest and sweetest water is every day carried up and is dissolved into vapour and rises to the upper region, where it is condensed again by the cold and so returns to the earth.'

Aristotle Fact 19: In his works on Meteorology, he talks about comets and meteors, and different types of weather, describing rainbows, thunder, lightning and winds. He discusses earthquakes which he believed were a result of underground winds.

Aristotle Fact 20: He died in Euboea in 322 BC, the same year that he'd arrived on the island. His death was a result of natural causes. The chief executor of his will was one of his students called Antipater. In his will, Aristotle requested that he be laid to rest next to his wife.

Influence & Legacy of Aristotle: Aristotle was an important philosopher who came from Greece. He offered his views and findings in various philosophical treatises. Aristotle made huge contributions towards a variety of subjects including physics and zoology, anatomy, embryology, geology, geography, astronomy, meteorology, literature and poetry. He based many of his findings on his own experiences and close observations.